

Anna-Drishti — PPT Context File

Source content for the six required slide fields, written to be copied straight into the deck.

Smart India Hackathon 2026 · Internal Hackathon Final · VIT

READ THIS BEFORE YOU BUILD SLIDES

This document answers exactly the six fields your template asks for, in order. Everything is written as slide-ready text — short sentences, no filler — so you can paste it in with light formatting rather than rewriting it under time pressure.

Two fields are not filled in for you: Problem Statement ID and, possibly, its exact title. Those come from your SIH portal registration, not from anything in this project. Filling them in wrong is worse than leaving them blank, so read the notes in those sections rather than guessing.

Project	Anna-Drishti — Seeing Inside the Grain Stack
Team	Chiranjib Dash (Lead) · Mohit Kumar · Vishal Vivek · Krittika Parida · Netra R · Devavrath Ramesh
Institution	Vellore Institute of Technology
Mentor	Prof. Bhabani Shankar Dey (Faculty ID 23278)

Not filled in here — this is not something this project can determine. The Problem Statement ID is assigned by the SIH portal when you register against an official published statement. It does not exist until you have selected one.

WHAT TO ACTUALLY DO

1. Check your team's SIH portal dashboard or registration confirmation email — the ID appears there once a Problem Statement has been selected.
2. If your internal round does not require mapping to an official national PS yet, write **“To be assigned”** or **“Student Innovation category — ID pending national round”** rather than leaving the field visually empty.
3. If several official PS statements look close to this project, pick the one whose title and description overlap most with grain storage, post-harvest loss, or agri-sensing — do not force-fit one that only partially matches.

DO NOT INVENT AN ID

A fabricated or guessed ID is worse than an honest placeholder. Judges and portals can check it in seconds, and being wrong here undermines trust in everything that follows on the slide.

IF YOU HAVE MATCHED AN OFFICIAL PS

Use its title **exactly as written** on the portal. Do not paraphrase it — judges compare the slide against the published wording.

IF THIS IS STUDENT INNOVATION, OR NO PS ID EXISTS YET

Use the project's own title, taken directly from the submitted proposal so it stays consistent across every document your team has produced this weekend:

SUGGESTED TITLE — COPY EXACTLY

Anna-Drishti — Seeing Inside the Grain Stack: Predictive Spoilage and Pest Intelligence for Indian Godowns

If the template wants a shorter one-line version for a title slide rather than a full subtitle, use:

SHORT FORM

Anna-Drishti — Predictive Spoilage Detection for Bagged Grain Storage

COPY EXACTLY

Agriculture, FoodTech & Rural Development

This is the theme stated directly in the submitted proposal, and it is a strong, accurate fit — not a stretch. Supporting points if the slide has room for a sub-line:

- Addresses **post-harvest loss**, a core Agriculture-theme problem — India's cereal losses were assessed at 3.89–5.92 % in the NABCONS study reported to Parliament in 2024.
- Targets the **World's Largest Grain Storage Plan in the Cooperative Sector** — over 500 PACS-level godowns under construction, none of them instrumented today. That is the Rural Development angle: capacity is being built faster than the expertise to monitor it.
- FoodTech fit: converts a manual, periodic inspection process into a continuous, predictive one — the same shift FoodTech themes typically reward.

RECOMMENDATION

Hardware — with a software decision engine as the interpretation layer on top of it.

WHY HARDWARE IS THE RIGHT ANSWER

The core novelty of the project is physical sensing, not the software alone:

- A perforated **probe-lance array** sampling interstitial air and vibration at three depths — this is the thing nothing else on the market does for bagged stacks specifically.
- A **single shared CO₂ analyser** multiplexed across ports via a rotary manifold — a genuine hardware engineering decision, not a software one.
- Contact piezoelectric and MEMS microphone sensing for insect detection.

The software — the digital twin, the decision engine, the comparison mode — exists to prove the *interpretation logic* works. It is real code, but it is in service of a hardware product. If your portal only allows one category, Hardware is the more accurate and more defensible choice; picking Software would undersell the actual proposal.

SAY THIS IF A JUDGE ASKS WHY YOU DEMOED MOSTLY SOFTWARE

“The sensing hardware is largely commodity components — the hard engineering problem is the interpretation logic that turns noisy readings into a trustworthy prediction. That is what we built and validated first. Hardware fabrication and lab-scale validation are the immediately following phase, and they are scoped and budgeted in our proposal.”

This is a recommendation based on the project’s own proposal, not a rule. Confirm against whatever categories your specific portal actually offers.

FIELD 5 Technologies to be Used

Split honestly into what exists in the proposal (the target system) and what has actually been built and tested for this internal round. **Keep this distinction on the slide** — judges reward teams who are precise about what is prototype versus plan far more than teams who blur it.

SOFTWARE & COMPUTATION

TECHNOLOGY	ROLE	STATUS
Vanilla JavaScript, HTML5, CSS	Entire digital-twin simulation and decision engine	Built
HTML5 Canvas (hand-rolled, no chart library)	Heatmap rendering, live probe charts, comparison timeline	Built
Explicit finite-difference numerical solver	Moisture, temperature and CO ₂ diffusion through the grain bed	Built
Modified Chung–Pfoest sorption isotherm (ASAE D245)	Converts measured humidity to water activity	Built
Cardinal-value predictive mycology model	Fungal growth-rate and mould-onset prediction	Built
Append-only hash-chained log	Tamper-evident condition and treatment record	Planned

Deliberately zero external dependencies and zero CDN references — the entire simulation runs fully offline in a single HTML file, by design, since rural godown connectivity cannot be assumed.

SENSING HARDWARE

TECHNOLOGY	ROLE	STATUS
ESP32 microcontroller	Sensor read-out and firmware	Built & flashed
Sensirion SHT3x (I ² C digital humidity/temperature)	Water-activity measurement, CRC-checked readings	Built & validated
SSD1306 OLED display	Live on-device readout	Built
NDIR CO ₂ analyser, single-unit + rotary manifold	Depth-resolved gas mapping across multiple ports from one instrument	Proposed, not yet sourced
Contact piezoelectric sensor + MEMS microphone	Acoustic detection of insect larvae feeding inside kernels	Proposed
Perforated probe lance (3 units, machined)	Non-destructive multi-depth sampling in a bag stack	Proposed

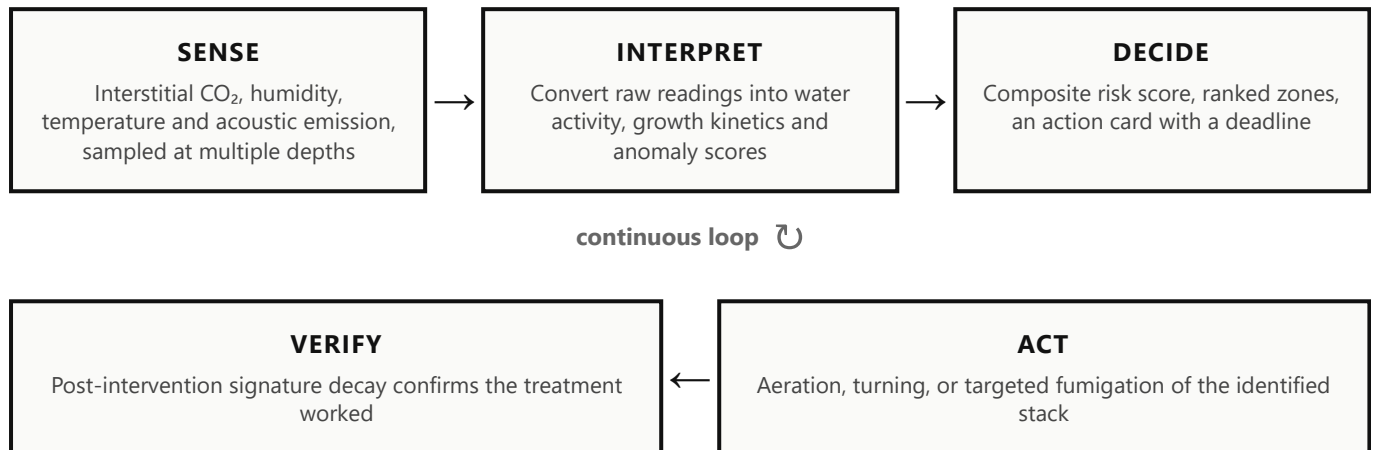
STANDARDS AND REFERENCE METHODS USED

- **ASAE D245** sorption-isotherm coefficients (shelled corn)

- FAO *Towards Integrated Commodity and Pest Management in Grain Storage* — inspection and detection guidance
- Published cardinal growth values for *Aspergillus flavus* (predictive mycology literature)
- Saturated-salt water-activity calibration standards (NaCl, K₂SO₄)

1 · THE CORE CONTROL LOOP

The system is architected as a closed loop, not a one-way dashboard. This is the diagram to put on your methodology slide:



This closes a loop that current practice leaves open at both ends — nothing today verifies that a treatment actually worked, so failures are typically only discovered when infestation returns.

2 · INTERPRETATION PIPELINE — PHYSICS TO BIOLOGY

The step that converts a raw sensor number into an actionable prediction, in the order it actually executes:

1. **Water activity** — measured interstitial relative humidity at equilibrium approximates grain water activity, via the sorption isotherm.
2. **Fungal growth kinetics** — water activity and temperature feed a cardinal-value growth model, giving days-to-onset for each zone.
3. **Respiration anomaly** — CO₂ is compared differentially against the other ports at the same depth on the same instrument, which cancels drift and slow ventilation changes.
4. **Insect classification** — acoustic streams are filtered for impulse rate and spectral signature, then matched against published species profiles.

3 · WORKING PROTOTYPE

This is not a mockup. It is a running, self-checking simulation with real finite-difference physics, a real sorption isotherm, and a real decision engine, verified against headless assertions before every session.

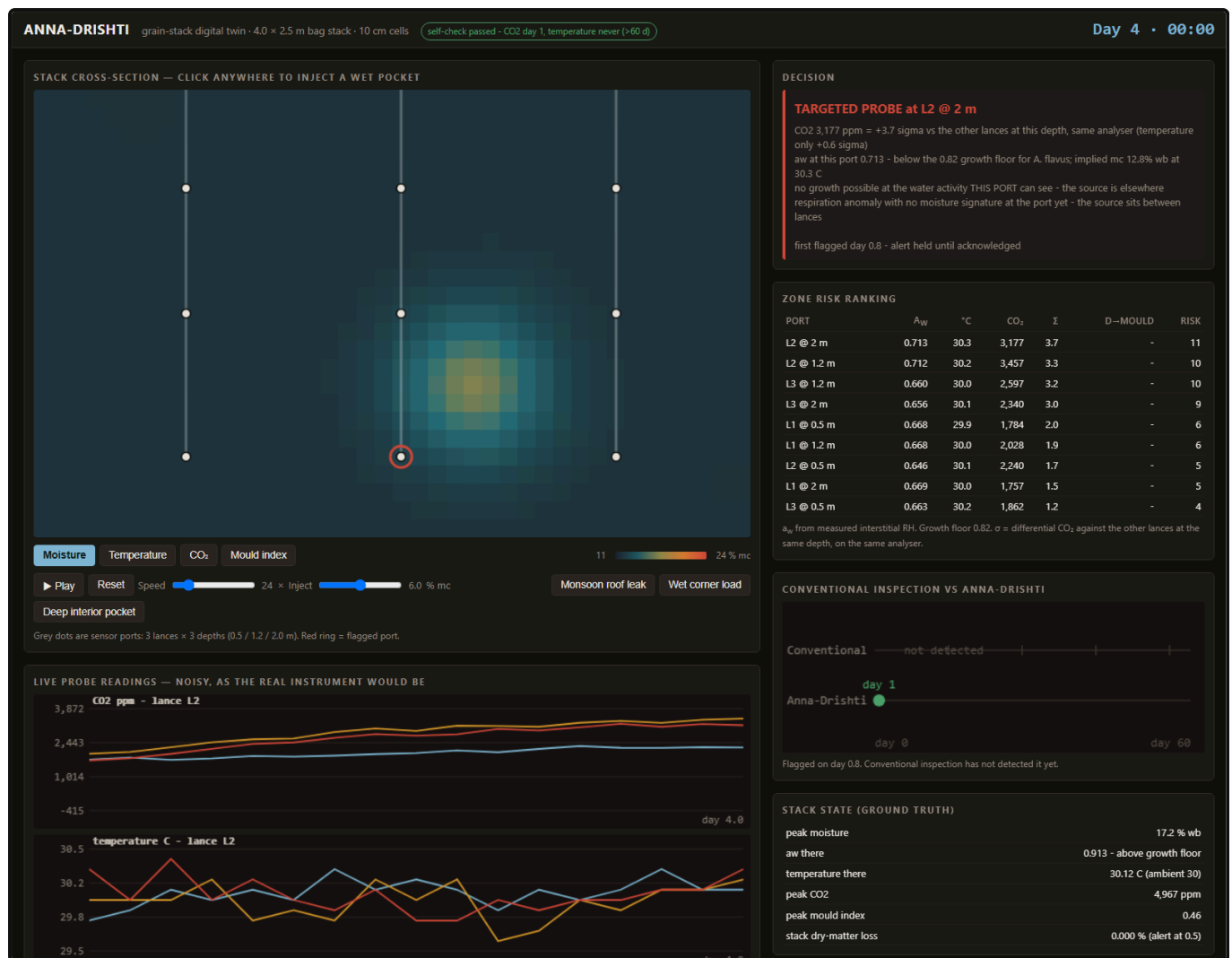


Figure 1. The working digital-twin prototype, four simulated days after a hotspot is injected 1.6 m into the stack. The decision engine has already fired a **Targeted Probe** alert (top right) naming the exact port, the evidence behind it, and a predicted deadline — while the conventional-inspection comparison (right column) shows the same event still undetected. The live CO₂ and temperature charts (bottom left) show the separation the whole system is built around: gas signal rising well before any temperature change appears.

4 · PHYSICAL VALIDATION — WHAT IS REAL, NOT SIMULATED

To avoid the project resting on simulation alone, a physical water-activity measurement was run in parallel using the sensing hardware actually built:

- Two sealed jars of maize, identical except for moisture content — one at storage moisture, one deliberately wetted.
- An SHT3x sensor, on the same ESP32 firmware built for the field probe, reads the headspace of each jar.
- The measured relative humidity is fed into the **same** decision-engine code used by the simulation — one source of truth for the science, not two.
- Result: the dry jar returns water activity below the 0.82 growth floor; the wet jar returns water activity above it, with a predicted mould-onset timeline.

STATE THIS BOUNDARY EXPLICITLY ON THE SLIDE

The jars demonstrate a **measured** water-activity difference through real hardware and real code. They do not yet demonstrate observed spoilage or a measured CO₂ signal — those require a longer incubation window and a CO₂ sensor not yet sourced. Being precise about this distinction is stronger than blurring it.

5 · IMPLEMENTATION STACK SUMMARY

One line, useful as a methodology-slide footer:

COPY EXACTLY

Sense (probe array) → Interpret (isotherm + growth kinetics, in-browser JavaScript) → Decide (composite risk scoring) → Act (targeted intervention) → Verify (post-treatment signature decay) — validated today against a working software prototype and a live water-activity measurement from grain sealed less than 24 hours ago.

Not a required field, but useful if the deck otherwise needs structuring from scratch. Each row maps to roughly one slide.

#	SLIDE	CORE CONTENT
1	Title	Project name, PS ID/title, theme, team
2	The problem	3.89–5.92 % post-harvest loss; interior of a stack is invisible; aflatoxin is irreversible
3	Why existing approaches fail	Spear sampling misses interior pockets; temperature cables are slow and built for silos, not bag stacks
4	The insight	CO ₂ moves through air, heat crawls through insulating grain — gas leads by roughly 15× over the same distance
5	Methodology	The Sense→Interpret→Decide→Act→Verify loop diagram from Field 6
6	Live demo / working prototype	Figure 1, or a live screen share of <code>index.html</code>
7	Physical validation	The jar experiment, water activity measured through real hardware
8	Comparison	Conventional inspection vs Anna-Drishti timeline, the headline detection-lag number
9	Technologies used	Field 5 tables, built vs planned
10	Cost & deployment	₹19,800 prototype budget, offline-first design, PACS-level target
11	Team & roadmap	Why this team, what Phase 2 looks like